

Module 2 – Practice Questions

Q1. In one sentence, what problem does a **perceptron** solve?

Answer : It helps to classify binary prediction where data can be separated with a linear decision boundary.

Q2. What are the three main components of a perceptron?

Answer : The inputs with weights, summation function plus bias and the activation function.

Q3. Is perceptron a classifier or a regressor? Explain briefly.

Answer : A perceptron is a classifier because it maps all input into a single output 0 or 1 ,where a regressor predicts continuous numeric values.

Q4. Why do we need to update weights in a perceptron?

Answer : Weights are updated to minimize error between the predicted output and the target output, making the model more accurate by finding the correct decision boundary.

Q5. What happens if we never update the weights during training?

Answer : If we never update weights the perceptron will make the same error again and again without learning anything.

Q6. Which two quantities decide **how much** the weight changes?

Answer : The error $(y - \hat{y})$ and the learning rate eta (η) decide how much the weight changes.

Q7. What does the term $(y - \hat{y})$ represent in perceptron learning?

Answer : $(y - \hat{y})$ represent the predicted error, here y is the actual label and $\hat{y}$ is the predicted label.

Q8. If the prediction is correct, will weights change? Why?

Answer : if we see carefully the weight update equation $W_{new} = W_{old} + \eta(y - \hat{y})$, here if the prediction is correct then the $(y - \hat{y})$ error becomes 0 ,for this $W_{new} = W_{old}$ no weight update occurs .

Q9. What role does the learning rate (η) play intuitively?

Answer : The learning rate eta (η) acts as a step size , means how fast or slow the model will adjust the weights in response to an error.

Q10. Write the AND gate truth table.

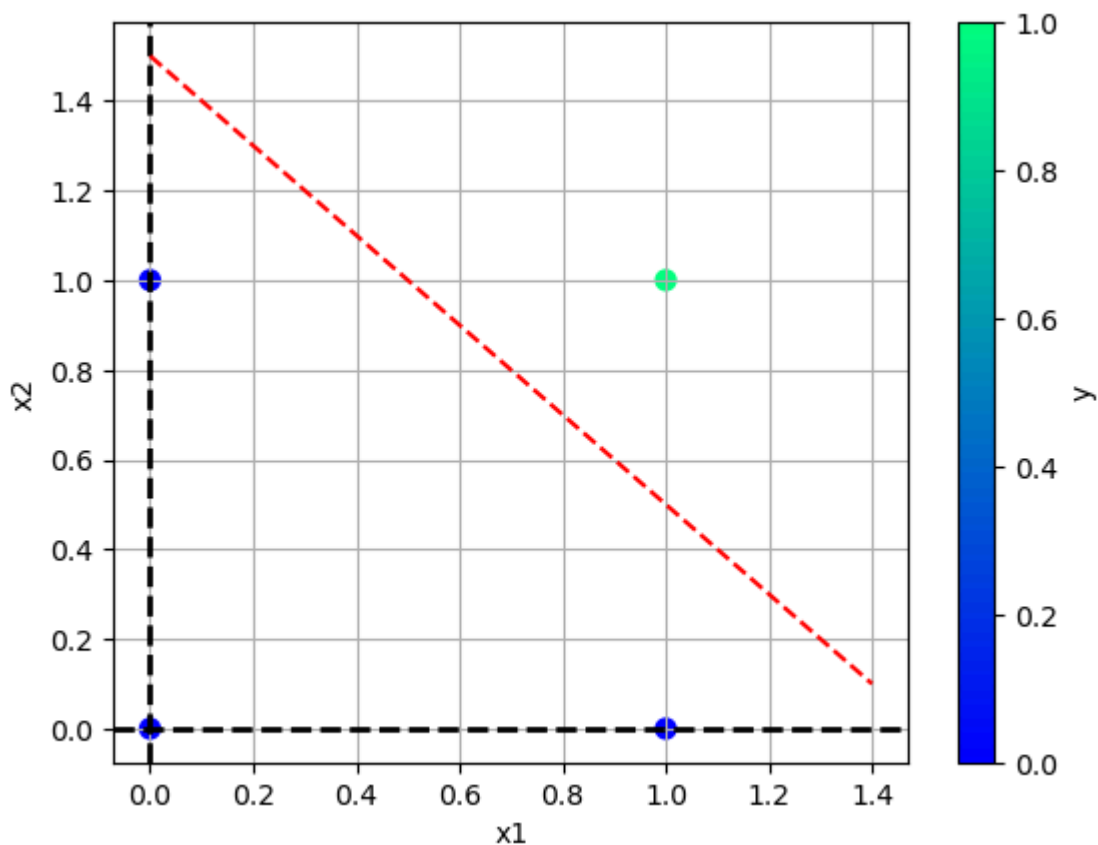
Answer : for $(0 \& 0 = 0)$, $(0 \& 1 = 0)$, $(1 \& 0 = 0)$, $(1 \& 1 = 1)$

Q11. Why is AND gate linearly separable?

Answer : Because we can draw a line in a 2D graph that can separate the output 0s and 1.

Q12. Draw a rough decision boundary that separates AND gate outputs.

Answer :



Q13. Why do we add a bias term in the perceptron?

Answer : Bias breaks the linearity , it shifts the decision boundary from the origin.

Q14. Why is bias often added **after summation**?

Answer : it acts as an offset to the weighted sum and helps to shift the threshold of the activation function.

Q15. Why do we initialize weights with small random values?

Answer : To break the symmetry so that the model can start learning without being biased or getting stuck.

Q16. What is the purpose of `np.dot(X, w)` in perceptron code?

Answer : It calculates the weighted sum of given inputs and weights before adding bias.

Q17. Why is matrix transpose used in weight updating?

Answer : The matrix is transposed to perform valid matrix multiplication.

Q18. What will happen if we remove the activation function?

Answer : Without the activation function the perceptron will output continuous linear numerical values like a linear regressor . It will lose the classification ability.

Q19. Why can OR gate be solved using a single perceptron?

Answer : The OR gate is linearly separable and for this, a decision boundary can be drawn , so a perceptron can solve the OR gate.

Q20. What is common between AND and OR gates from a geometry view?

Answer : In a 2D graph the AND and OR gate data points can be clearly separable according to their groups with a decision boundary.

Q21. Does XOR need more features or more layers? Why?

Answer : Yes XOR needs more features or more layers because it does not show a linearly separable decision boundary like AND and OR , which means a non-linear relationship, which can be solved through multi-layer perceptrons.

Q22. What does *linearly separable* mean?

Answer : It means when data can be perfectly separable by a linear decision boundary like a straight line, a plane or a hyperplane.

Q23. Why can't a single straight line separate XOR data?

Answer : In an XOR gate the output 1 can be derived from both (0,1) and (1,0) inputs when 0 also comes from both (0,0) and (1,1). In the graph these positions are connected diagonally across from each other so a single decision boundary can never be drawn.

Q24. What change is required to solve XOR successfully?

Answer : To solve XOR successfully we need to use a Multi-layer perceptron .

Q25. List two strengths of a perceptron.

Answer :

1. It is very fast to train and requires little memory.
2. If the data is linearly separable then it guarantees a solution in a finite number of steps.

Q26. List two limitations of a perceptron.

Answer :

1. A single perceptron only works to solve linearly separable data
2. It can only classify binary probability (0/1)

Q27. When should we avoid using a single-layer perceptron?

Answer : When the data shows non linear behaviour then we need to avoid using a single-layer perceptron.

Mini Challenge (Very Easy)

Q28.

Given:

- $x_1 = 1, x_2 = 1$
- weights = [1, 1]
- bias = -1.5

Will the perceptron output be **1 or 0**?

(Show reasoning in one line)

Answer : We know , $Z = x_1 w_1 + x_2 w_2 + \dots + x_n w_n + b$

So , $(1*1)+(1*1)+(-1.5) = 0.5$

As per sigmoid function $z > 0$ here , that is why the perceptron will output 1 here .